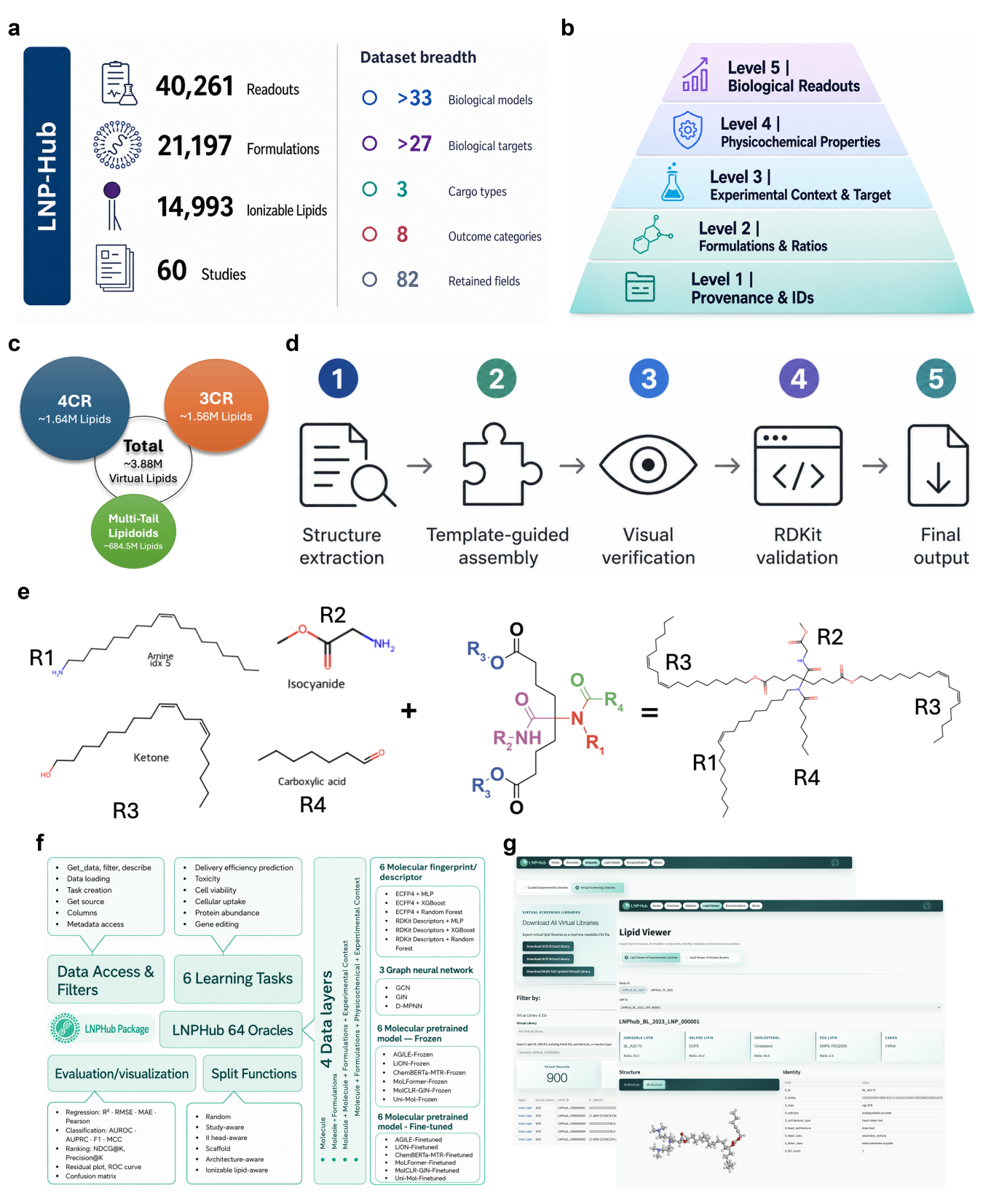

a
LNP-Hub
40,261 Readouts
21,197 Formulations
14,993 Ionizable Lipids
60 Studies
Dataset breadth
>33 Biological models
>27 Biological targets
3 Cargo types
8 Outcome categories
82 Retained fields
b
Level 5 | Biological Readouts
Level 4 | Physicochemical Properties
Level 3 | Experimental Context & Target
Level 2 | Formulations & Ratios
Level 1 | Provenance & IDs
c
4CR ~1.64M Lipids
3CR ~1.56M Lipids
Total ~3.88M Virtual Lipids
Multi-Tail Lipidoids ~684.5M Lipids
d
1 Structure extraction
2 Template-guided assembly
3 Visual verification
4 RDKit validation
5 Final output
e
R1
Amine idx 5
R2
Isocyanide
Ketone
R3
Carboxylic acid
R4
f
Get_data, filter, describe
Data loading
Task creation
Get source
Columns
Metadata access
Data Access & Filters
Delivery efficiency prediction
Toxicity
Cell viability
Cellular uptake
Protein abundance
Gene editing
6 Learning Tasks
LNPHub Package
LNPHub 64 Oracles
Evaluation/visualization
Split Functions
Regression: R² · RMSE · MAE · Pearson
Classification: AUROC · AUPRC · F1 · MCC
Ranking: NDCG@K, Precision@K
Residual plot, ROC curve
Confusion matrix
Random
Study-aware
II head-aware
Scaffold
Architecture-aware
Ionizable lipid-aware
4 Data layers
Molecule
Molecule + Formulations
Molecule + Molecule + Formulations + Experimental Context
Molecule + Formulations + Physicochemical + Experimental Context
6 Molecular fingerprint/ descriptor
ECFP4 + MLP
ECFP4 + XGBoost
ECFP4 + Random Forest
RDKit Descriptors + MLP
RDKit Descriptors + XGBoost
RDKit Descriptors + Random Forest
3 Graph neural network
GCN
GIN
D-MPNN
6 Molecular pretrained model — Frozen
AGILE-Frozen
LiON-Frozen
ChemBERTa-MTR-Frozen
MolFormer-Frozen
MolCLR-GIN-Frozen
Uni-Mol-Frozen
6 Molecular pretrained model - Fine-tuned
AGILE-Finetuned
LiON-Finetuned
ChemBERTa-MTR-Finetuned
MolFormer-Finetuned
MolCLR-GIN-Finetuned
Uni-Mol-Finetuned
g
Lipid Viewer
Download All Virtual Libraries
LNPhub_BL_2023_LNP_000001
900